

Simulated annealing placement

Simulated annealing jogs one cell at a time under an HPWL cost

The idea

- Propose a small axis move, accept improvements always
- Keep the best iterate, not only the final temperature
- Fix the seed so your golden stays stable across runs

Pseudocode

- Placement annealing jogs one cell per move under HPWL cost, accepts by Metropolis
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Seed forty-two with sixty moves yields best HPWL near forty-nine point six from fifty-two
- Lock the seed so goldens stay stable

Algorithm sketch — try these

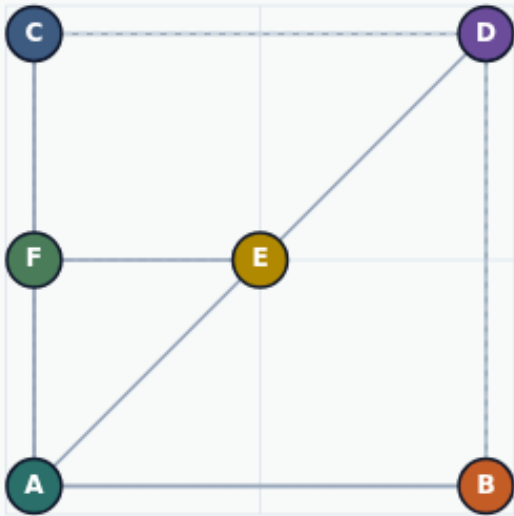
```
INPUT: positions, seed, moves, T0
OUTPUT: best positions + best HPWL
for i in 1..moves:
    jog one cell on one axis
    accept if  $\Delta\text{HPWL} < 0$  or  $\text{rand} < e^{(-\Delta/T)}$ 
    keep best; cool T
GOLDEN seed=42, 60 moves  $\rightarrow$  best $\approx$ 49.6
accepted $\approx$ 44 rejected $\approx$ 16
```

Anneal from HPWL fifty-two

SA PLACEMENT

Anneal from HPWL fifty-two

Step 1 / 5 · starter-sa · module02-07-sa-placement



Explanation

Simulated annealing jogs one cell at a time under an HPWL cost. Seed forty-two and sixty moves on the starter —deterministic teaching run.

- Random axis jogs
- Cost = total HPWL
- seed = 42 · moves = 60

Start HPWL: 52
 $T_0 \approx 3$ · cool ≈ 0.94

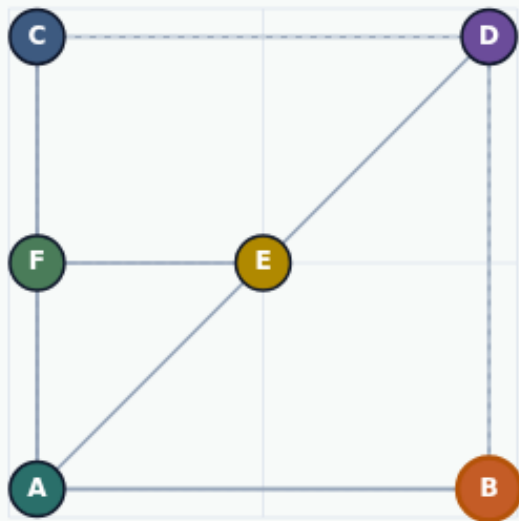
Anneal from HPWL fifty-two

Accept improvements; maybe worsenings

SA PLACEMENT

Accept improvements; maybe worsenings

Step 2 / 5 · accept-reject · module02-07-sa-placement



Explanation

Improvements always accept. Worsenings accept with probability $\exp(-\Delta/T)$. Temperature cools each move so late stages get pickier.

- $\Delta \leq 0 \rightarrow$ accept
- Else accept with $e^{(-\Delta/T)}$
- Keep the best iterate

Accepted: 44

Rejected: 16

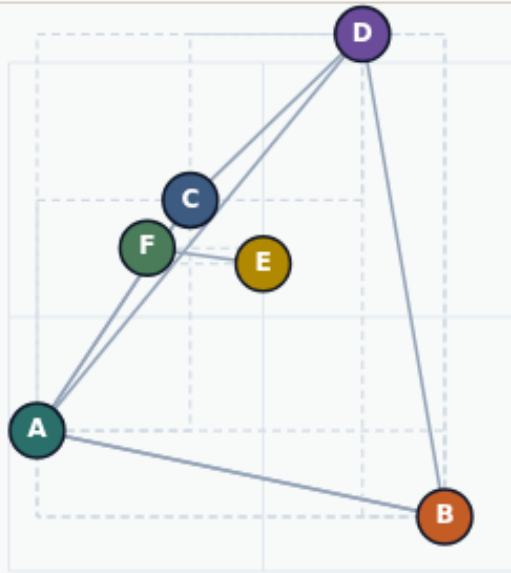
Accept improvements; maybe worsenings

Best HPWL near forty-nine point six

SA PLACEMENT

Best HPWL near forty-nine point six

Step 3 / 5 · best-not-final · module02-07-sa-placement



Explanation

With seed forty-two the best HPWL lands near forty-nine point six—modest gain over fifty-two. Report best, not only the final temperature.

- Best $\neq$ last iterate
- Modest SA gain on this tiny netlist
- Force still wins pure WL here

saHpwlAfter: 49.6
Measured best: 49.6
accepted/rejected: 44/16

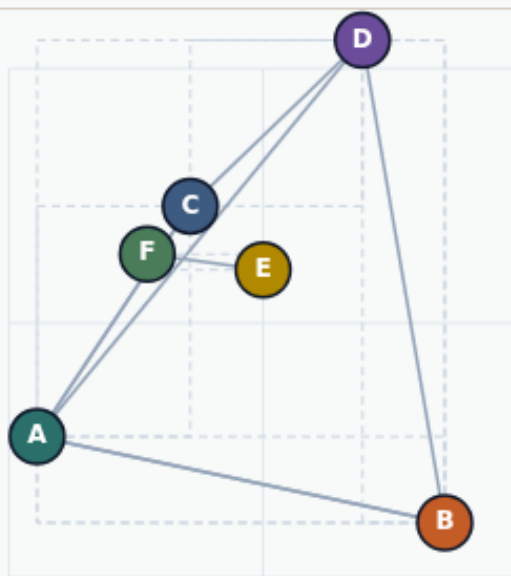
Best HPWL near forty-nine point six

Fix the seed for goldens

SA PLACEMENT

Fix the seed for goldens

Step 4 / 5 · seed-lock · module02-07-sa-placement



Explanation

Change the seed and accept/reject counts drift. Teaching labs lock seed forty-two so forty-nine point six stays a stable challenge.

- Deterministic PRNG (mulberry32)
- seed 42 → ≈ 49.6 best
- 44 accept / 16 reject

saHpwlAfter: 49.6
saAccepted: 44
saRejected: 16

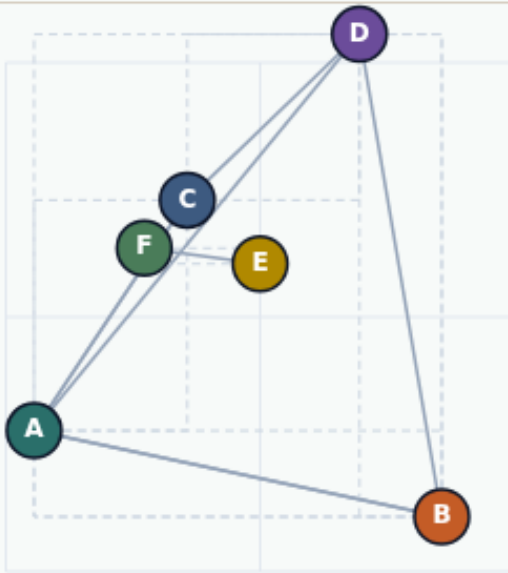
Fix the seed for goldens

SA is a stochastic local search

SA PLACEMENT

SA is a stochastic local search

Step 5 / 5 · takeaway · module02-07-sa-placement



Explanation

Propose, score, accept or reject, cool, and remember the best. On this instance expect about forty-nine point six—not a force-style collapse.

- Best HPWL ≈ 49.6
- Lock seed 42
- Next: density bins

saHpwlAfter: 49.6

SA is a stochastic local search

Browser lab track

- In the browser lab track, open the **sa-placement** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse `tiny_place.json`, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module